

Prime Quadruplets as Complete Cycles in $(\mathbb{Z}/12\mathbb{Z})^\times$ and Their Interpretation via Dirichlet Characters

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Abstract

We study the structure of prime quadruplets $(p, p+2, p+6, p+8)$ via residue classes modulo 12. We show that all admissible quadruplets correspond to complete cycles in the multiplicative group $(\mathbb{Z}/12\mathbb{Z})^\times$. This classification can be expressed naturally in terms of Dirichlet characters modulo 12.

We further compare this local arithmetic structure with global spectral properties associated with the Riemann zeta function. Numerical experiments indicate that while quadruplet structure is fully explained by local congruence constraints, no direct correlation with spectral spacing statistics is observed.

1 Introduction

Prime constellations are constrained by congruence conditions modulo small integers. In this work, we analyze prime quadruplets

$$(p, p+2, p+6, p+8)$$

using residue classes modulo 12.

We introduce a four-state classification of primes and show that quadruplets correspond to complete cycles in this system.

2 Residue Classes Modulo 12

For primes $p > 3$, one has

$$p \equiv 1, 5, 7, 11 \pmod{12}.$$

Define:

$$E := 1, \quad A := 5, \quad B := 7, \quad C := 11.$$

Proposition 1. *The set $(\mathbb{Z}/12\mathbb{Z})^\times = \{1, 5, 7, 11\}$ forms a group isomorphic to $C_2 \times C_2$.*

Proof. Direct computation shows closure and that each element has order 2. Hence the group is the Klein four group. $\square$

3 Dirichlet Character Representation

Let χ_1, χ_2 be two independent Dirichlet characters modulo 12. Then every prime $p > 3$ can be mapped to

$$p \mapsto (\chi_1(p), \chi_2(p)) \in \{\pm 1\}^2.$$

Proposition 2. *The assignment*

$$E \leftrightarrow (+1, +1), \quad A \leftrightarrow (-1, +1), \quad B \leftrightarrow (+1, -1), \quad C \leftrightarrow (-1, -1)$$

defines a bijection between residue classes and character values.

4 Prime Quadruplets

Theorem 1. *Let $(p, p+2, p+6, p+8)$ be a prime quadruplet with $p > 3$. Then the residues modulo 12 form a cyclic permutation of (E, A, B, C) .*

Proof. Checking all admissible residues modulo 12 shows that only $p \equiv 5$ or $11 \pmod{12}$ can yield quadruplets. Explicit evaluation gives:

$$5 \rightarrow 7 \rightarrow 11 \rightarrow 1, \quad 11 \rightarrow 1 \rightarrow 5 \rightarrow 7.$$

Thus all quadruplets traverse all four residue classes exactly once. □

Prime quadruplets are minimal complete cycles in $(\mathbb{Z}/12\mathbb{Z})^\times$.

5 Dirichlet L-functions

For a Dirichlet character χ , define

$$L(s, \chi) = \prod_p (1 - \chi(p)p^{-s})^{-1}.$$

The Riemann zeta function is

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1}.$$

5.1 Interpretation

The zeta function aggregates all primes, while $L(s, \chi)$ isolates arithmetic information associated with residue classes.

6 Numerical Experiments

We analyze a dataset of Riemann zeros γ_n .

6.1 Spacing Distribution

Define

$$s_n = \gamma_{n+1} - \gamma_n.$$

Normalized spacings follow the expected GUE-type distribution.

6.2 Quadruplet Analysis

Quadruplets were computed up to a cutoff N . Observations:

- Only two cyclic patterns occur (corresponding to starting residues 5 and 11)
- All quadruplets realize full cycles in $(\mathbb{Z}/12\mathbb{Z})^\times$

6.3 Correlation Test

We compare quadruplet locations with spacing data.

Proposition 3. *No statistically significant correlation between quadruplet positions and normalized zeta spacing was detected.*

7 Discussion

The results indicate a clear separation:

- Local structure: governed by congruence constraints modulo 12
- Global structure: governed by spectral properties of $\zeta(s)$

Prime quadruplets arise purely from local admissibility conditions.

8 Conclusion

We have shown:

Prime quadruplets $\leftrightarrow$ complete cycles in $(\mathbb{Z}/12\mathbb{Z})^\times$

and:

EABC classification $\leftrightarrow$ Dirichlet character decomposition.

This provides a minimal and exact structural explanation of quadruplets.